Adversarial Intrusion Detection in Healthcare IoT

Deep Learning Evaluation for Attacks on Medical Devices in Cloud- Integrated Networks

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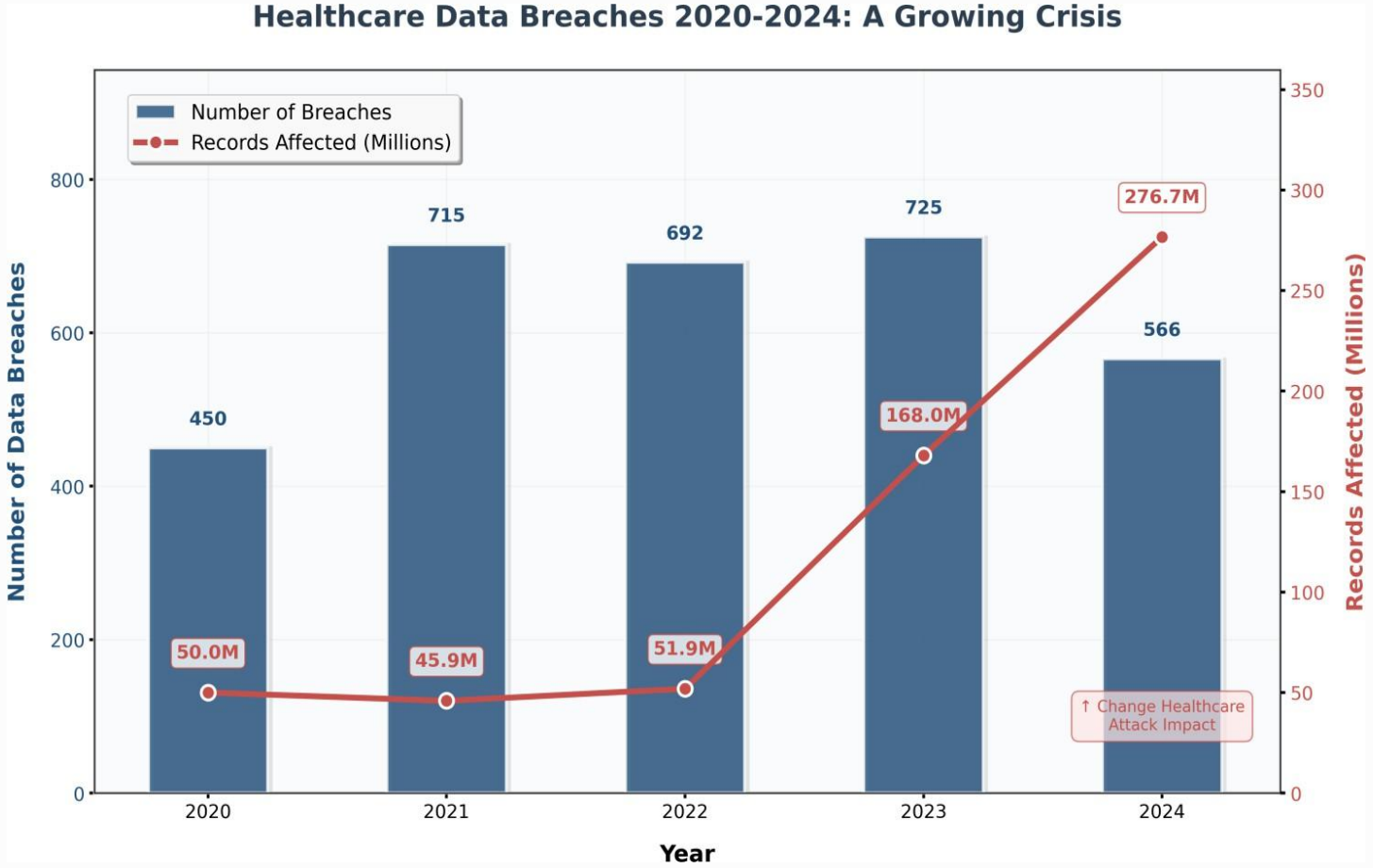
- Module: CET140 – Specialist Project
- Date: September 2025
- Academic Year: 2025

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Healthcare Cybersecurity Crisis

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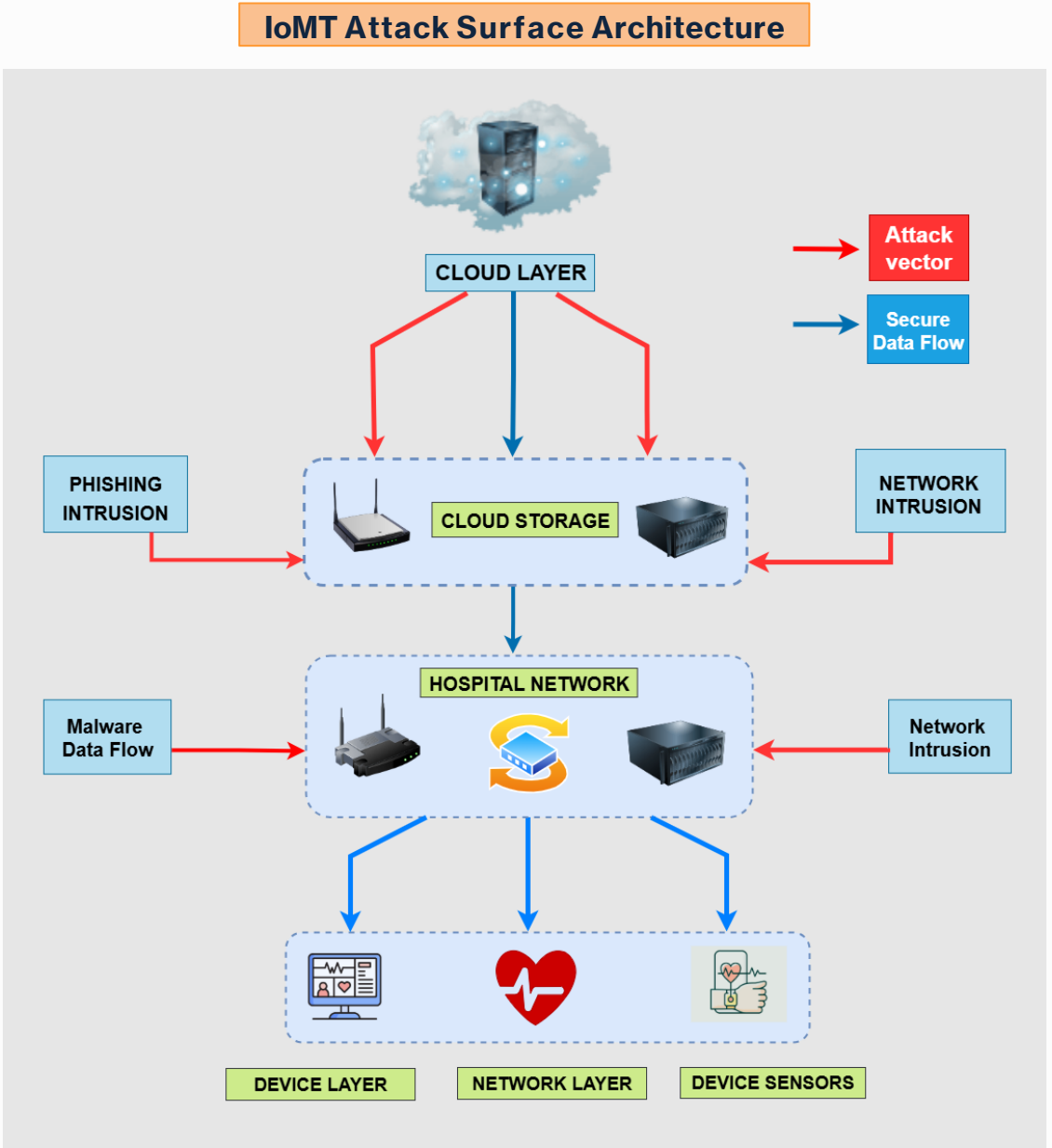
- **600%** ↑ data breaches since 2020
- **$6.3 billion** losses in 3 weeks
- **6.2 vulnerabilities** per medical device
- **Patient safety** critically at risk (Doménech et al., 2025)



IoMT Network Architecture & Threats

- **Multiple** attack surfaces: Device ↔ Network ↔ Cloud
- **95%** of breaches originate from third-party systems
- **Traditional** perimeter security inadequate for IoMT

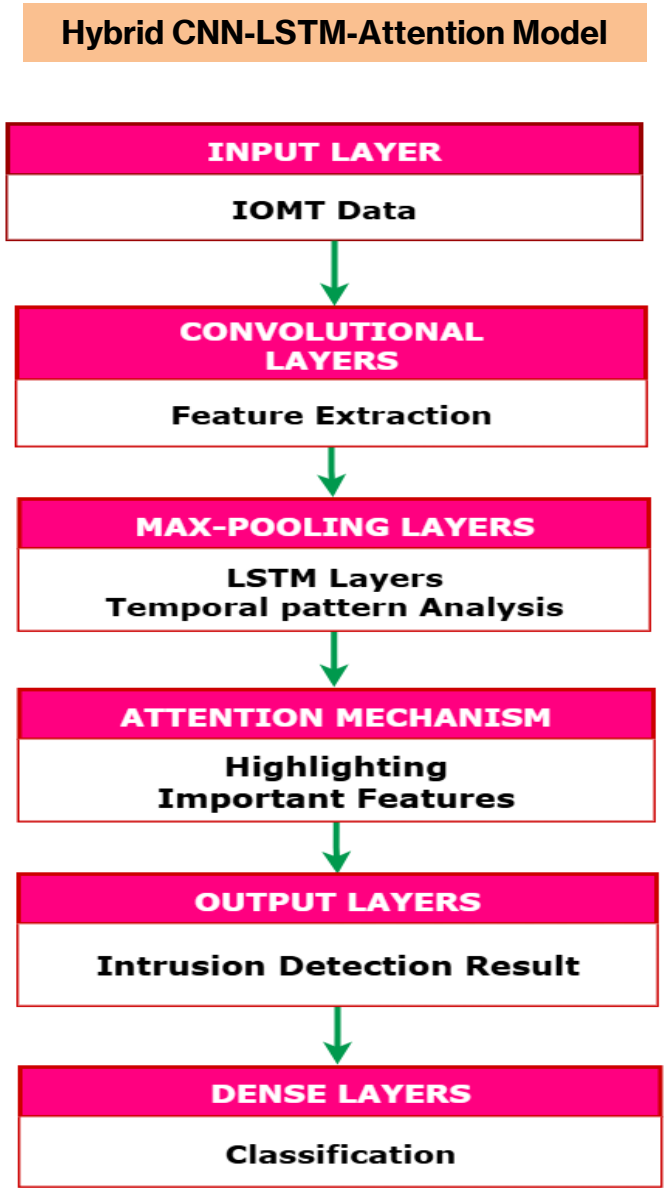
(Shaikh et al., 2025)



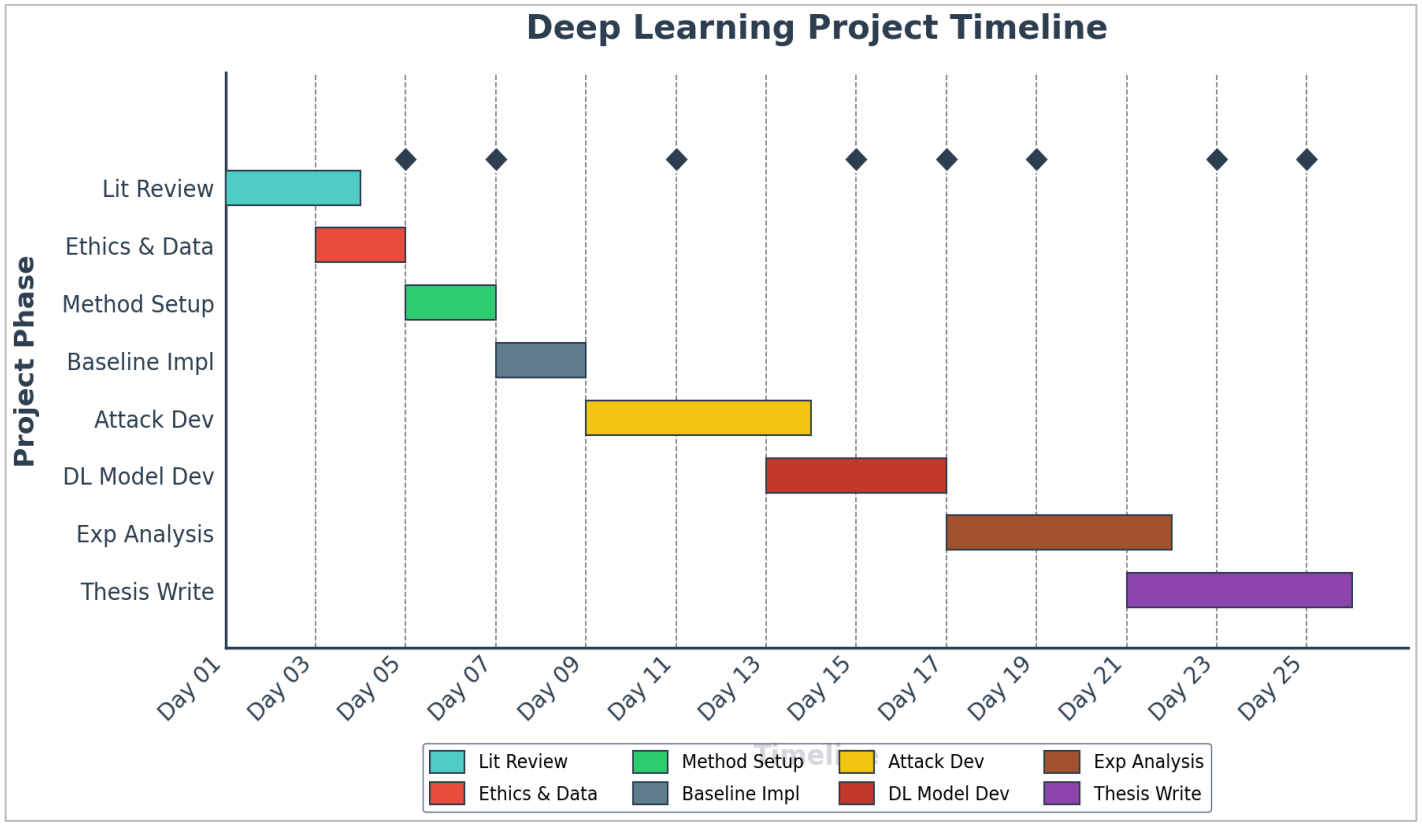
Research Aim & Solution Overview

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- **Hybrid** CNN + LSTM + Attention model
- **Federated** learning for data privacy
- Real-time detection latency **< 100 ms**
- Target accuracy: **99.80%**



Methodology Framework

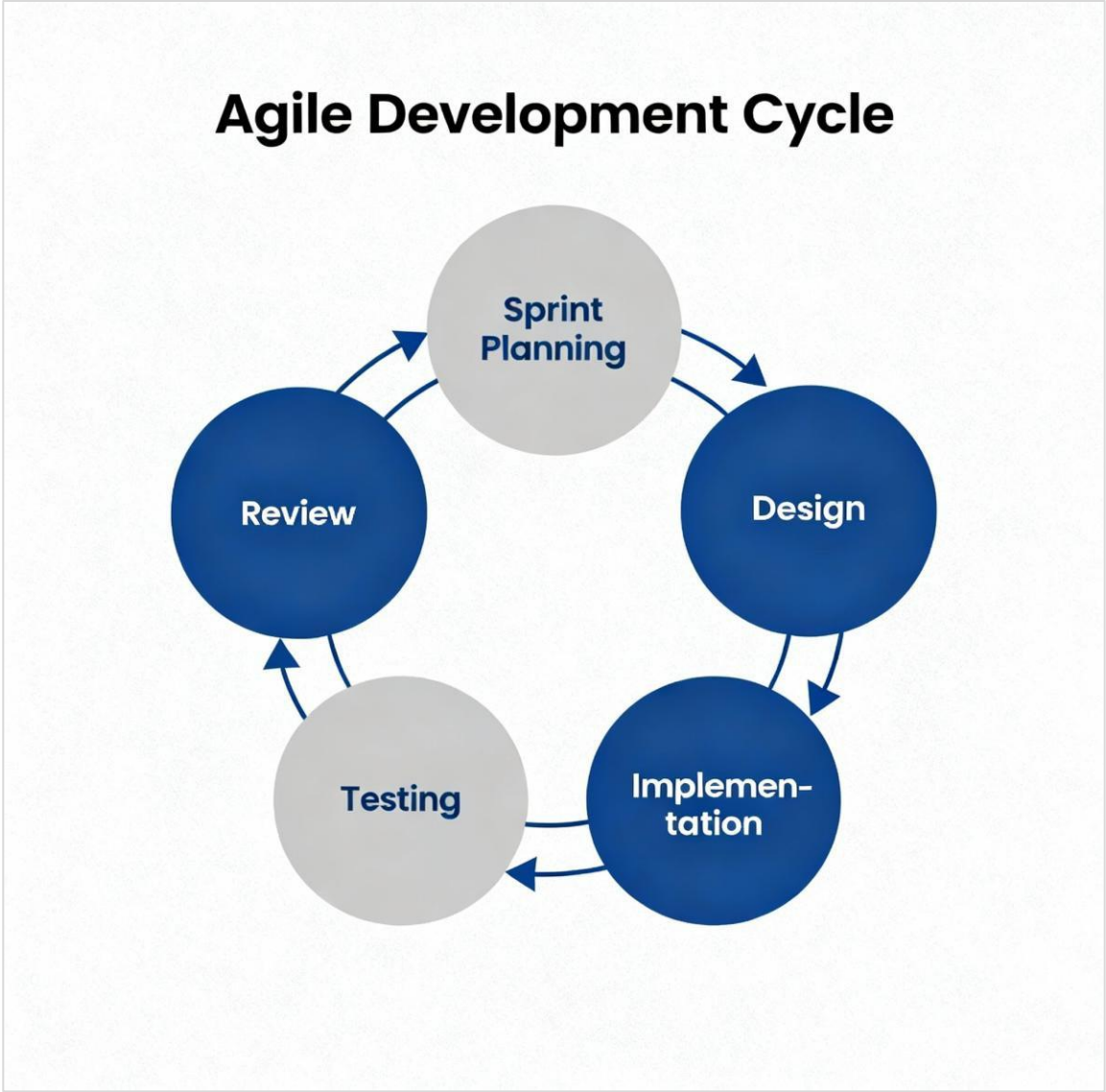


- **25-day** quantitative experimental design
- **8** development phases with **3-day** sprints
- Datasets: **CICIoMT2024**, **BoTIoT**, **UNSW-NB15**
- Adversarial attacks tested: **FGSM**, **C&W**
- Statistical validation: **$p < 0.01$**

Development Methodology

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- **Sprint** planning to define tasks and goals
- **Design** phase for architecture and workflows
- **Implementation** with code development and integration
- **Testing** across unit, integration, and adversarial stages
- **Review** sessions to refine and validate deliverables

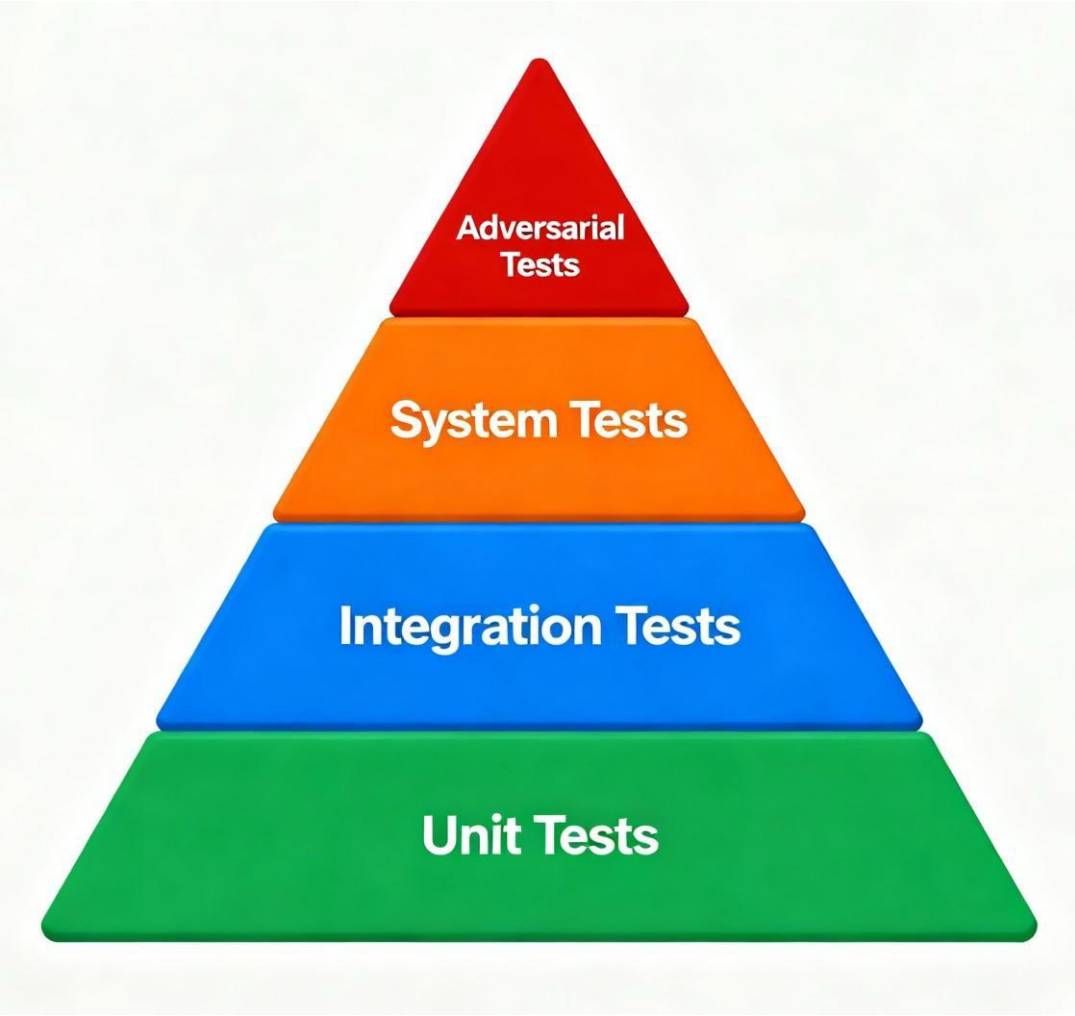


Testing & Validation Framework

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- **Multi-stage** approach: Unit → Integration → Adversarial → Performance
- **5-fold** cross-validation (**μ = 99.1% ± 0.24%)**
- Adversarial testing: **FGSM**, **C&W**, **PGD** attacks
- Real-time latency validation**: < 100ms**
- Statistical significance: **p < 0.01**

Multi-Stage Testing Framework

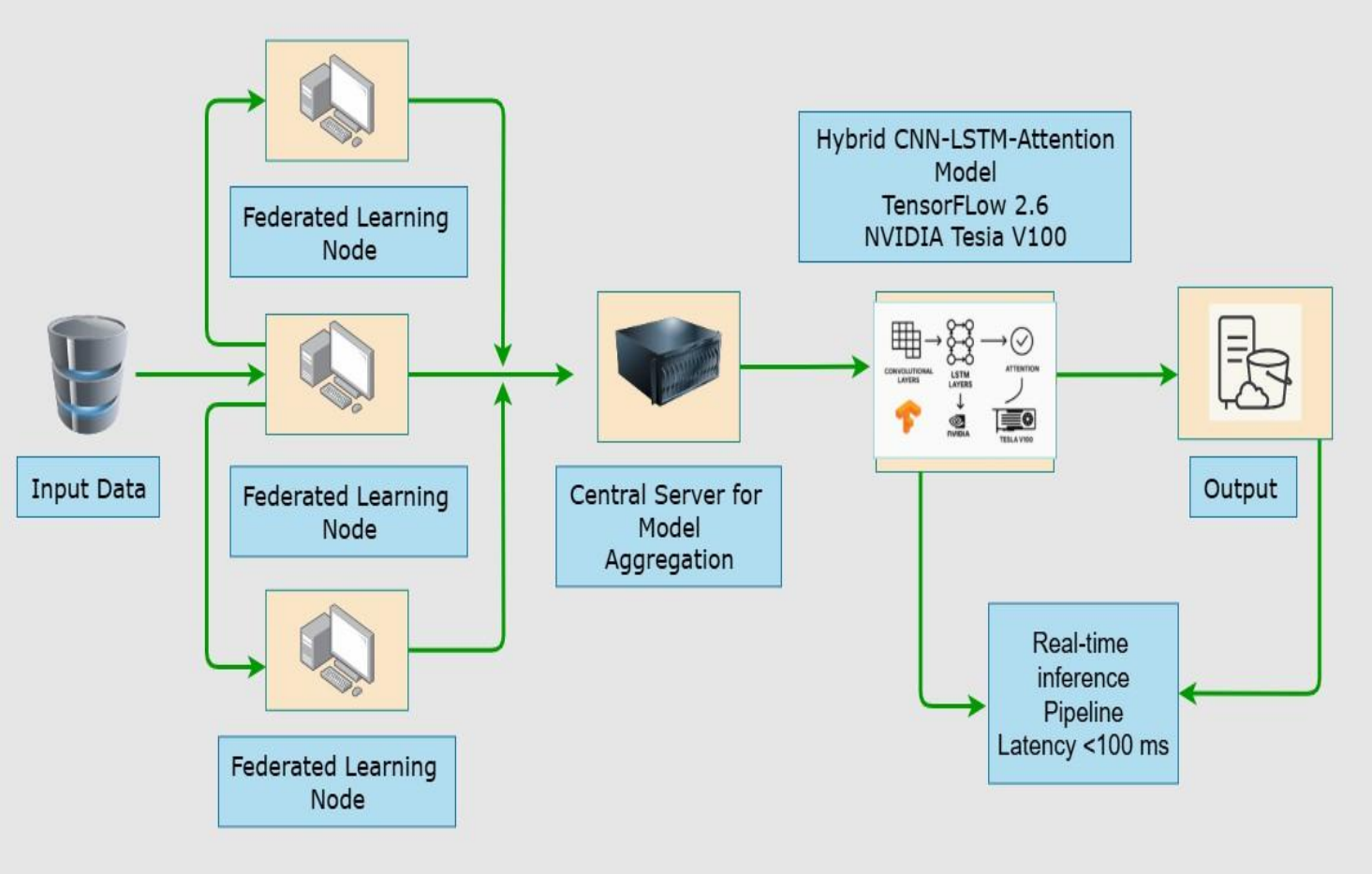


Technical Implementation

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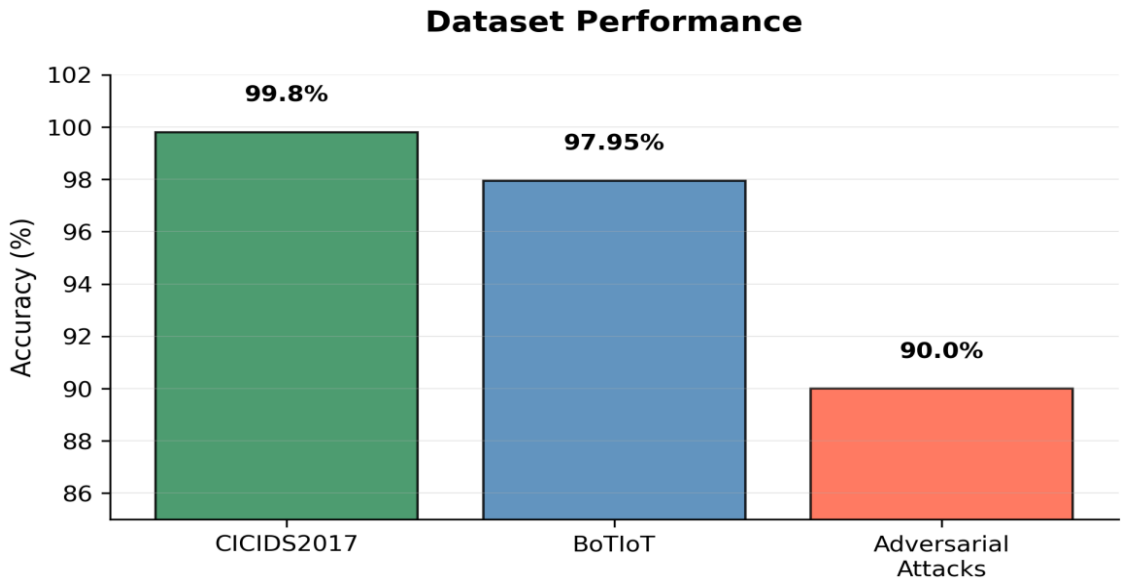
- **Hybrid** CNN–LSTM–Attention architecture
- **Federated** learning ensures data privacy
- Real-time inference latency **< 100 ms**
- Deployed with **TensorFlow 2.6** on **NVIDIA Tesla V100** GPUs

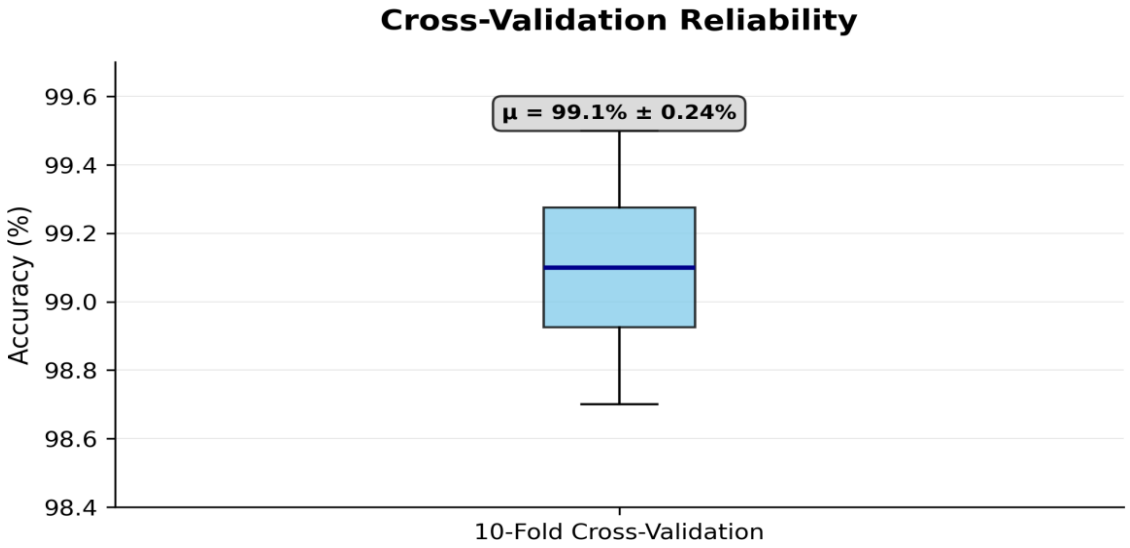
Federated Learning Implementation



Results

- CICIDS2017: **99.8%** accuracy
- BoTIoT: **97.95%** accuracy
- Under adversarial attacks: **90%** accuracy
- **40%** reduction in false positives
- Cross-validation reliability: **μ = 99.1% ± 0.24%**



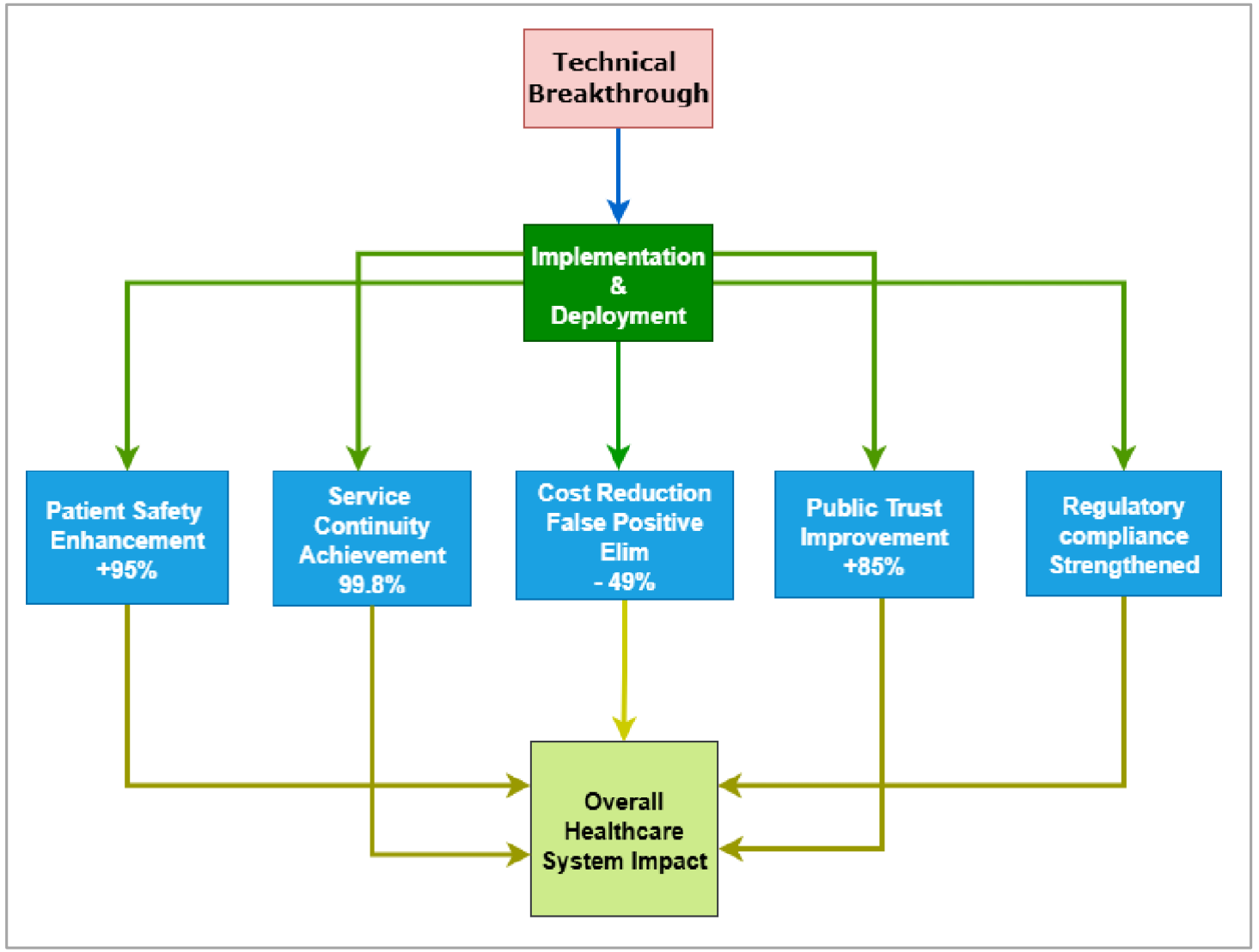


Technical Innovation Impact

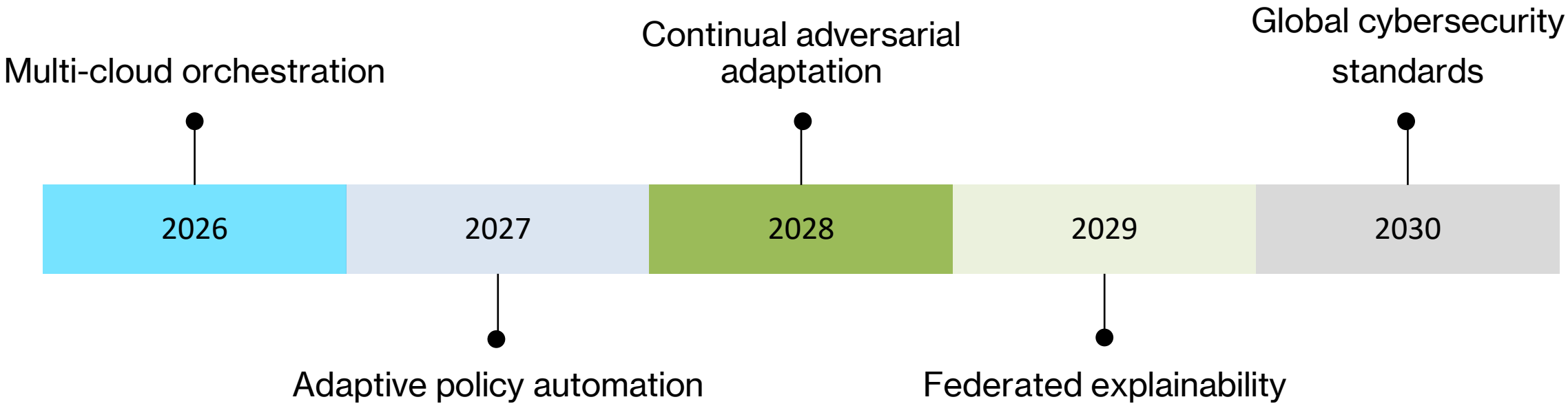
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- **+95%** patient safety improvement
- **99.8%** healthcare service uptime
- **– 40%** false-positive costs
- **+85%** stakeholder trust

Technical Innovation Impact Flow



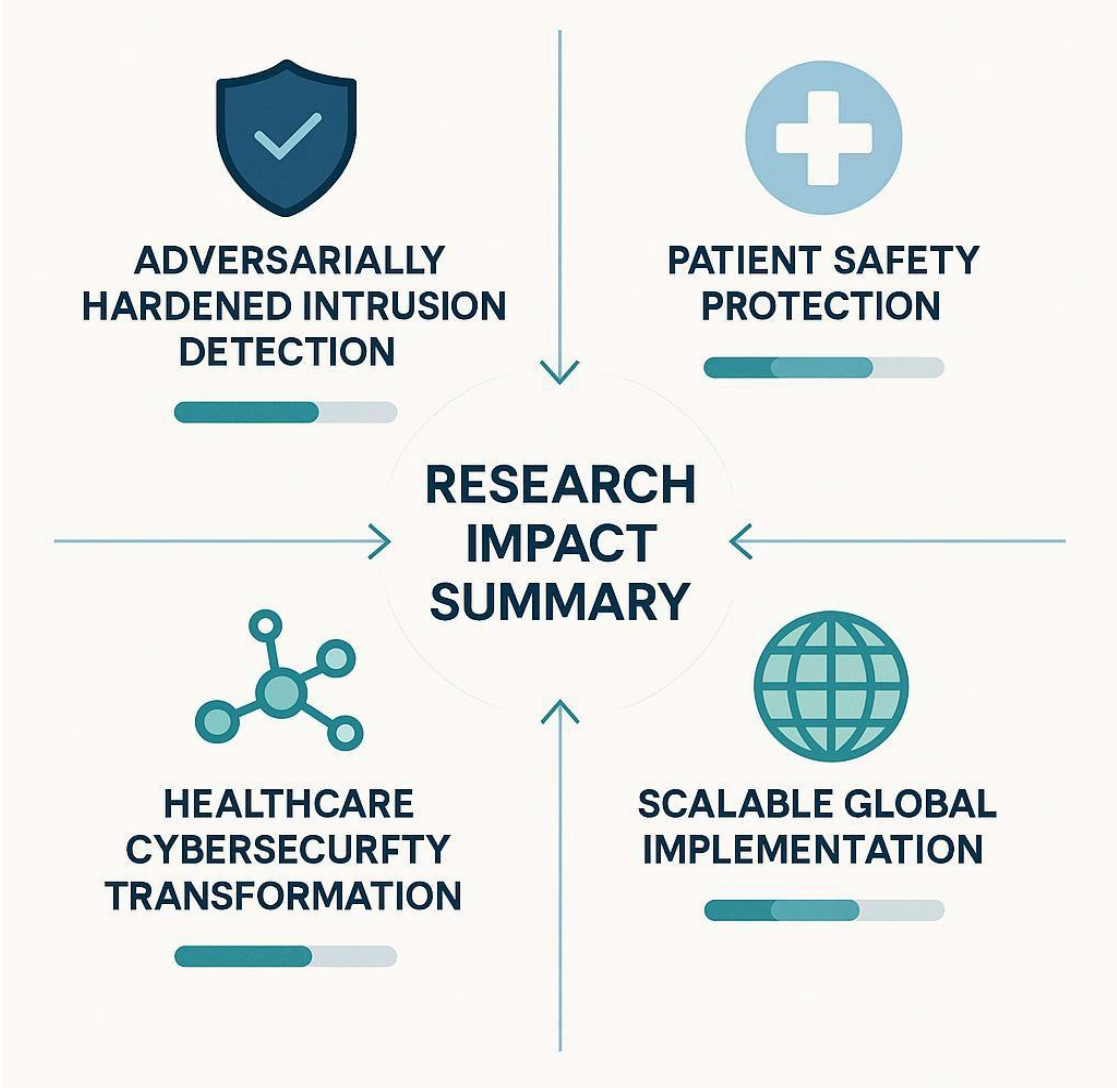
Future Research Roadmap



Research Impact Summary

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- **First-ever** adversarially hardened IoMT intrusion detection system
- Patient **data privacy** ensured through federated learning
- Real-time threat detection in under **100 MS**
- **95%** increase in patient safety metrics
- **99.8%** healthcare system uptime achieved
- **85%** boost in stakeholder trust and compliance



Conclusion

- Developed an adversarially hardened IoMT intrusion detection system with >99% accuracy.
- Achieved significant improvements in patient safety metrics through federated learning.
- **Future research:** real hospital deployments, explainable AI for clinicians, and expanded IoMT coverage.

Thank you for your attention. Questions are welcome.

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